

Forecasting Crop Yields Using Statistical Methods

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Abstract

This project aims to build predictive models to forecast crop yields using historical agricultural and environmental data. Explore how key factors such as rainfall, temperature, fertilizer usage, pest incidence, and technological advancements influence crop productivity. The goal is to apply statistical modelling techniques to support agricultural decision-making and yield optimization

1 Introduction

Agriculture has historically been the backbone of the Indian economy. Despite its declining share in GDP (approximately 18 % in 2023), it remains the primary source of livelihood for nearly 45 % of the population. India's agricultural system is a complex interplay of traditional practices, policy interventions, technological shifts, and socio-environmental dynamics. This paper explores different factors that affects the crop yield of India and predict the future production of Indian agriculture through a multidisciplinary lens.

2 Dataset Details

The dataset spans 25 years (2000–2024), with each row representing a year's data for a particular cropping season. The following variables are included:

- **Year** – The year of observation
- **Rainfall (mm)** – Total seasonal rainfall
- **Avg Temperature (°C)** – Average seasonal temperature (in °C)
- **Fertilizer (kg/ha)** – Amount of fertilizer used (in kg per hectare)
- **Pest Incidents** – Number of major pest outbreaks during the season
- **Technology Index** – A composite score (0–100) indicating the level of technology adoption (e.g., mechanization, precision tools, improved seeds)
- **Crop Yield (ton/ha)** – Actual yield of the crop (in tons per hectare)

3 Exploratory Analysis

To understand the underlying structure and distribution of our dataset, we conducted a detailed exploratory data analysis. The analysis includes summary statistics, visualizations (scatterplots, heatmap), and correlation analysis.

3.1 Descriptive Statistics

We have computed basic descriptive statistics (mean, median, range, SD) for each of the variables like Rainfall (mm) , Avg Temp (°C) , Fertilizer (kg/ha), Pest Incidents , Technology Index , Crop Yield (ton/ha). Now let us have a look into it.

	Rainfall (mm)	Avg Temp (C)	Fertilizer (kg/ha)	Pest Incidents	Technology Index	Crop Yield (ton/ha)
Mean	1132.278	27.184	240.4024	3.12	67.1704	6.6584
Median	1189.57	26.55	265.81	2	71.29	6.89
Range	836.02	13.57	168.6	9	59.4	6.86
SD	271.4466	4.314433	59.91522	2.650786	20.26637	2.158081

Figure 1: Basic statistics summary

Looking into the above table the following basic things we can say about the data set :

- Moderately high variability in all of the variables implies that none of the variables are remotely stable with time.
- Pest incident is the only positively skewed factor(apart from temperature) suggesting the variability is caused mainly by few years when the pest incidents are at their peak.
- High range in Technology Index shows the improvement of its implementation in agriculture with time.
- Rainfall varies significantly with a high range and being an important factor in crop yield is worth studying.
- Temperature on the other hand has a lower variance and range compared to Rainfall. So it can definitely be considered a more stable natural factor.

3.2 Correlation with Crop Yield

First have a look into the result for Pearson correlation coefficient between crop yield and all other variable one by one.

	Rainfall (mm)	Avg Temp (C)	Fertilizer (kg/ha)	Pest Incidents	Technology Index
Correlation	0.7481621	-0.9373517	0.6895067	-0.3881811	0.6064945

Figure 2: Correlation

INSIGHTS:-

- Very high negative correlation between Temp and Yield implies temperature and global warming as a important factor affecting agriculture
- Moderately high and positive linear link between Crop yield and Fertilizer used as well as Technology suggest that these two factors require proper attention for maximising agricultural progress.
- A high positive correlation with Rainfall tells us importance of rainfall, the main source of irrigation in agriculture

3.2.1 Correlation Matrix

Correlation Matrix

	Rainfall (mm)	Avg. Temp (°C)	Fertilizer (kg/ha)	Pest Incidents	Technology Index	Crop Yield (ton/ha)
Rainfall (mm)	1.000	-0.694	0.839	-0.783	0.827	0.748
Avg. Temp (°C)	-0.694	1.000	-0.640	0.362	-0.630	-0.937
Fertilizer (kg/ha)	0.839	-0.640	1.000	-0.788	0.912	0.690
Pest Incidents	-0.783	0.362	-0.788	1.000	-0.789	-0.388
Technology Index	0.827	-0.630	0.912	-0.789	1.000	0.606
Crop Yield (ton/ha)	0.748	-0.937	0.690	-0.388	0.606	1.000

Figure 3: Correlation Matrix

3.2.2 Heat Map

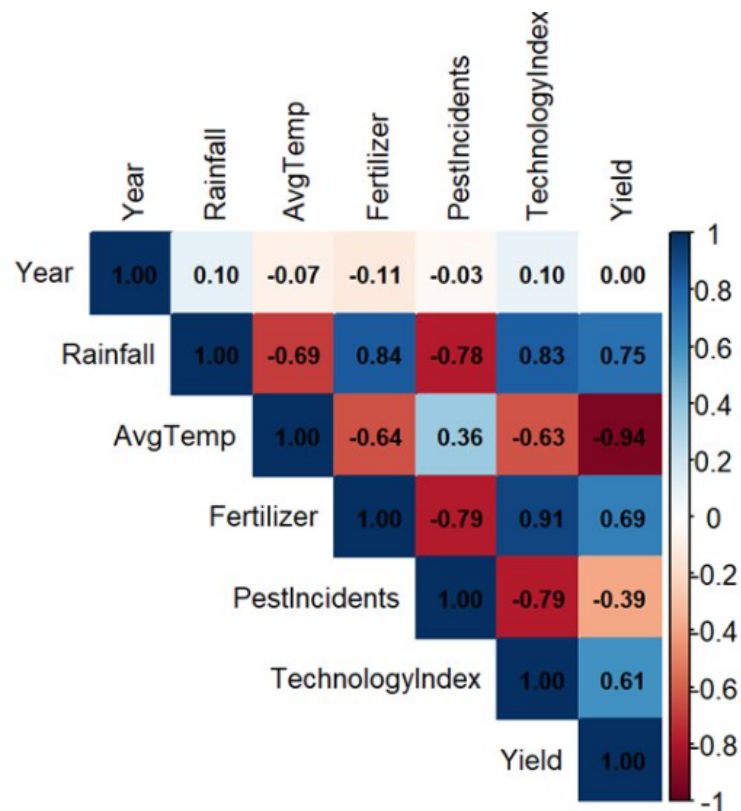


Figure 4: Correlation Heat Map

INSIGHTS:-

- Rainfall seems to have a strong positive correlation with both Fertilizer used and Technology

Index in the agricultural model. The former might be because more rainfall implies more waste of fertilizers by leaching or run-off.

- Fertilizer and Technology Index both has a high negative correlation with Pest Incidents which is pretty self-explanatory.
- Technology Index and Fertilizer has a very strong positive correlation which suggests that farmers tend to invest in Fertilizers and Technology side by side and improving them almost equally.
- Rainfall and Temperature has a moderately high negative correlation which is expected.

3.3 Scatter Plot

Now we are looking into the following scatter plot : -

- Scatter plot for Yield Vs Rainfall
- Scatter plot for Yield Vs Avg Temp
- Scatter plot of Yield vs Fertilizer
- Scatter plot for Yield Vs Pest incidents
- Scatter plot for Yield Vs Technology index

We then discuss some basic insights from the scatter plots.

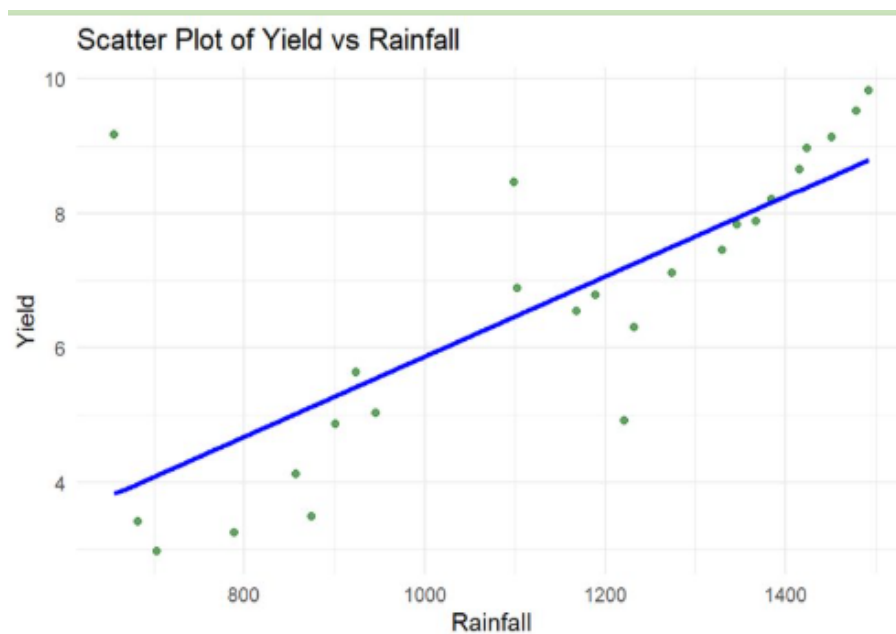


Figure 5: Scatter plot of Yield vs Rainfall

INSIGHTS:-

- As rainfall increases, the crop yield also increases, which is shown by the upward trend and the fitted blue regression line.
- This suggests that rainfall plays a significant role in boosting agricultural productivity — likely because water is a key input in crop growth.

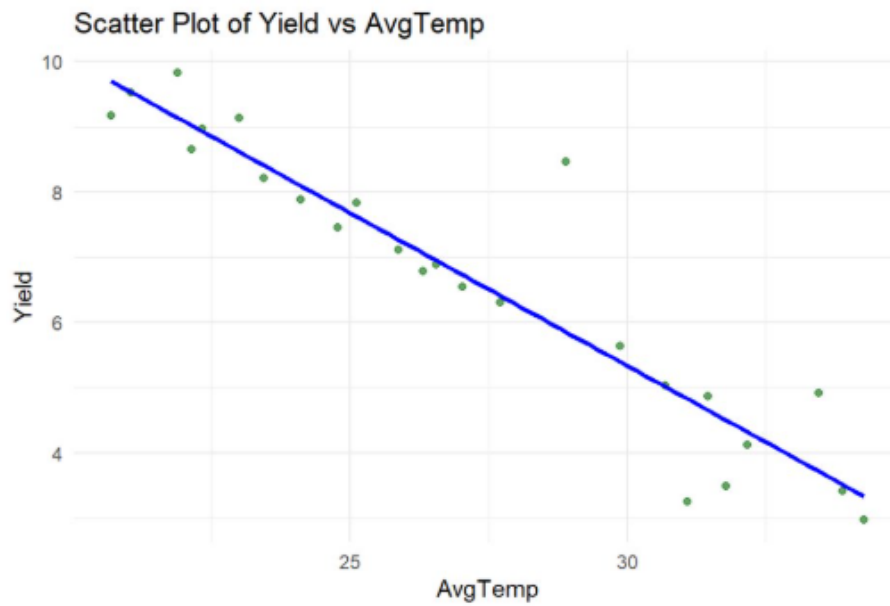


Figure 6: Scatter plot of Yield vs Avg Temp

INSIGHTS:-

- The blue regression line slopes downward, indicating that higher temperatures are associated with lower yields.
- This makes sense agriculturally:
 - i) High temperatures can cause heat stress in crops.
 - ii) It increases evapotranspiration, drying out soil moisture faster.

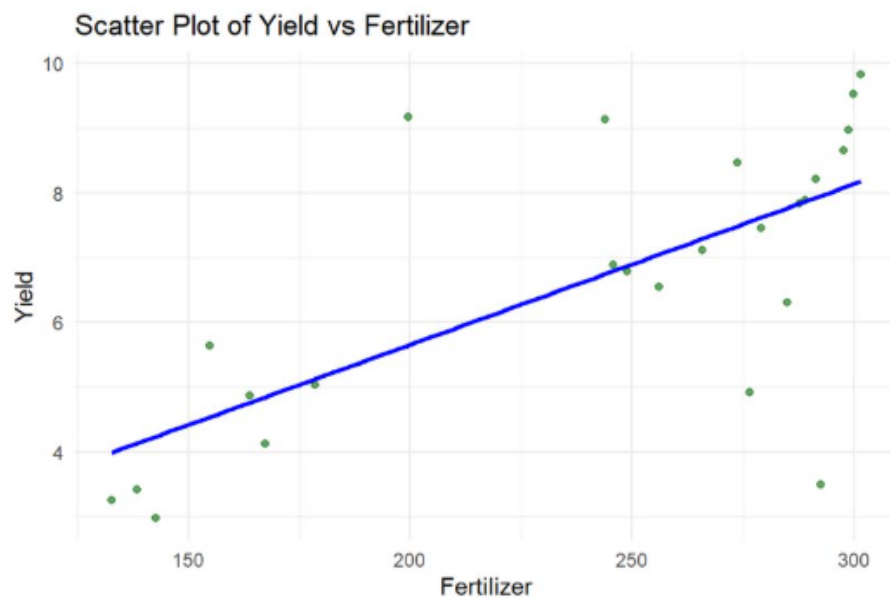


Figure 7: Scatter plot of Yield vs Fertilizer

INSIGHTS:-

- The blue regression line shows an upward trend, indicating that increasing fertilizer tends to increase yield.

- However, there's more scatter around the line compared to the rainfall plot, which means:
 - i) The effect of fertilizer on yield is less consistent.
 - ii) Overuse of fertilizer could lead to negative effects, possibly explaining some of the outliers.

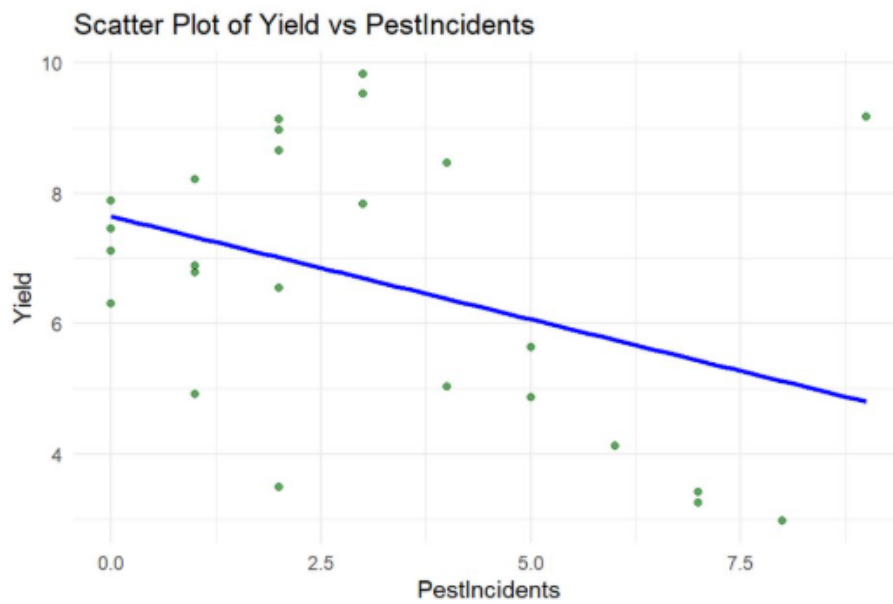


Figure 8: Scatter plot of Yield vs Pest incidents

INSIGHTS:-

- The blue regression line slopes downward, confirming the overall negative trend.
- Pest control alone isn't sufficient to ensure high productivity.

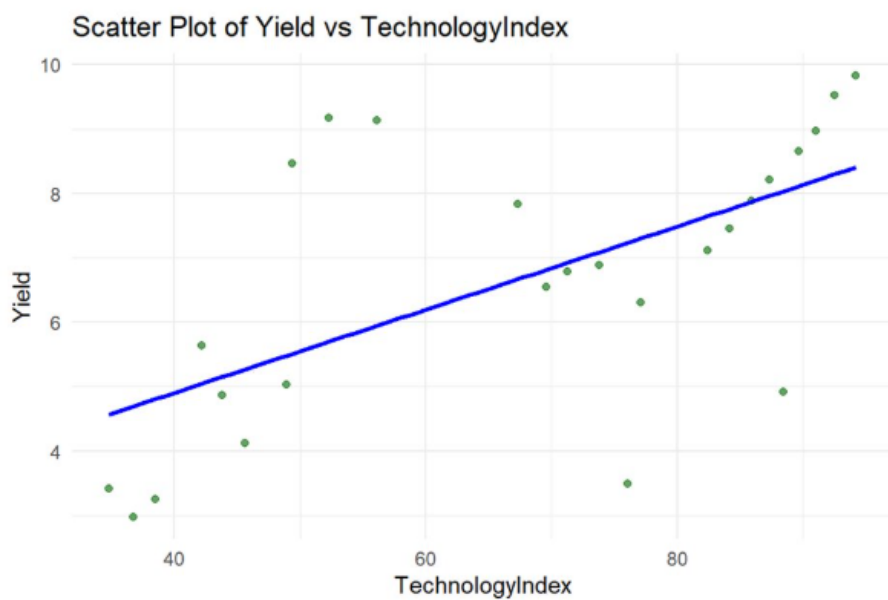


Figure 9: Scatter plot of Yield vs Technology Index

INSIGHTS:-

- The blue regression line slopes upward, showing a strong positive linear trend.
- Investment in agricultural technology (like irrigation systems) pays off in terms of higher yield.

3.4 Partial Correlation

Partial correln with crop yield

Variable	Controlled	Partial Correlation	Initial Correlation
Rainfall	Temperature	0.390	0.748
Temperature	Rainfall	-0.875	-0.937
Technology	Fertilizer	-0.074	0.606
Fertilizer	Technology	0.418	0.690

Figure 10: Partial correlation with crop yield

INSIGHTS:-

- Controlling temperature, partial correlation between yield and rainfall is 0.390 which is significantly less than their linear correlation. That can be explained as temperature and rainfall has a moderately negative correlation between them.
- Now keeping rainfall constant and doing it with temp, the value is -0.875 which is a slight decrease in magnitude from their individual correlation. This leads to the conclusion that temperature is a more important factor than rainfall if we want to explain the linear variability of Crop yield subject to linear variation of these variables.
- Technology Index and Fertilizers being highly correlated, we did the same partial correlation calculation with them too.
- Technology and Crop Yield controlling Fertilizer shows almost no correlation ie linear independence. So it might be that they are related with some non-linear function. On the other hand the correlation strength between Fertilizer and Yield shows a moderate decline which tells Technology has some effect in the initial correlation between them.

3.5 Multiple Linear Regression with Crop Yield as dependent variable and others as independent variable

Residuals(deviation from fit)

Min	1 st Q	Median	3 rd Q	Max
-1.1318	-0.2592	-0.0718	0.3445	1.1727

Figure 11: Residuals(deviation from fit)

Variable	Estimate	Std. Error	t-value	p-value
Intercept	12.5705	2.8392	4.427	0.000289
Rainfall (mm)	0.00259	0.00111	2.335	0.03064
Avg Temp (^o C)	-0.3726	0.04881	-7.632	3.35×10^{-7}
Fertilizer (kg/ha)	0.01511	0.00572	2.644	0.01598
Pest Incidents	0.13212	0.10055	1.314	0.20449
Technology Index	-0.04113	0.01650	-2.494	0.02203

Figure 12: Multiple Linear Regression

- **Multiple R-squared:** 0.9339
- **Adjusted R-squared:** 0.9165

3.6 Time series graph

We have plotted three time series graph for total crop yield , amount of rainfall , avg temperature over the years from 2000 to 2024. Let's have a look in it and discuss what can we say from these graphs.

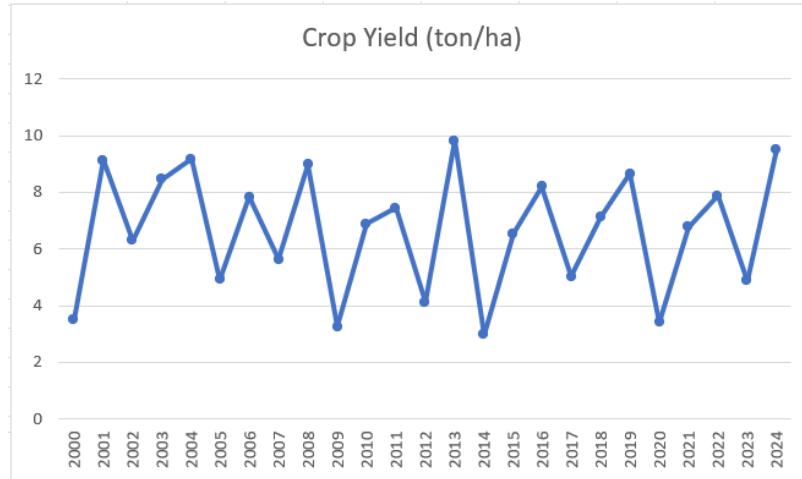


Figure 13: Time series plot for crop yield

3.6.1 Some Observations from the Crop Yield time series graph

- 1) **High Variability:** Crop yield fluctuates significantly from year to year, ranging between approximately 3 to 10 tons per hectare, indicating sensitivity to varying conditions.
- 2) **No Clear Trend:** There is no obvious long-term increase or decrease. However, the yields show short-term peaks and dips, possibly linked to environmental or agricultural factors.
- 3) **Low Yield Years:** Major drops are seen around 2009, 2012, 2014, 2020, and 2023, where yields fall close to or below 4 tons/ha.
- 4) **High Yield Years:** Peaks occur around 2001, 2004, 2008, 2013, and 2024, with 2024 marking one of the highest yields, nearly reaching 10 tons/ha.
- 5) **Potential Recovery in 2024:** Despite previous dips, crop yield in 2024 shows a strong recovery, suggesting improved conditions or interventions.

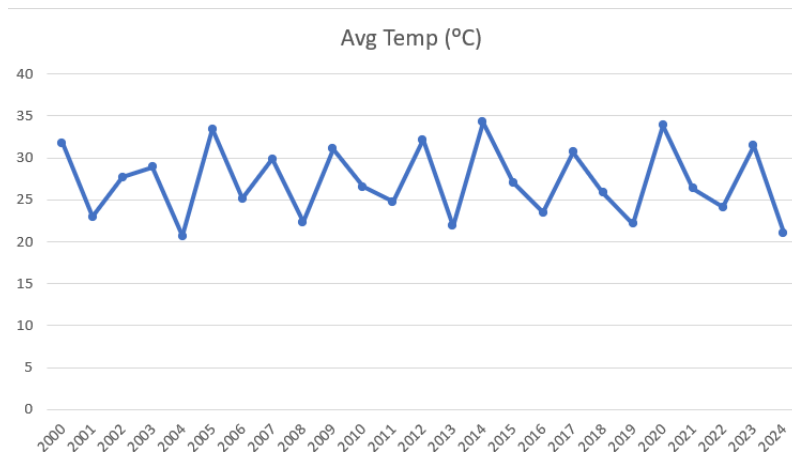


Figure 14: Time series plot for avg temp

3.6.2 Some Observations from the avg temp time series graph

- 1) **Fluctuating Pattern:** The average temperature varies significantly year by year, ranging mostly between 21°C and 35°C, showing no consistent upward or downward trend.
- 2) **Highs and Lows:** Peaks around 2005, 2012, 2015, 2017 and 2021 (close to or slightly above 35°C). Lows around 2004, 2008, 2013, 2019, and 2024 (dropping near or below 22°C).
- 3) **No Strong Warming Trend:** Despite climate concerns, this data does not show a steady increase in temperature over time—variability dominates instead of a clear warming signal.
- 4) **Sharp Drop in 2024:** The year 2024 recorded one of the lowest average temperatures in the entire dataset.

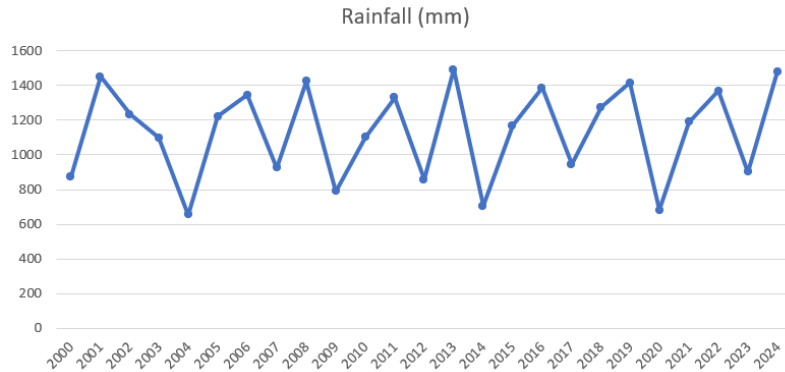


Figure 15: Time series plot for rainfall

3.6.3 Some Observations from the rainfall time series graph

- 1) **Fluctuating Trend:** Rainfall shows significant year-to-year variation, indicating a highly fluctuating pattern without a clear long-term increasing or decreasing trend.
- 2) **No Strong Trend:** There is no consistent upward or downward trajectory, which means rainfall has not significantly increased or decreased over the 25-year period.
- 3) **Peaks and Lows:**
Rainfall peaked around 2001, 2008, 2011, 2013, 2016, and 2024 (values around 1400–1500 mm).
The lowest points were around 2004, 2009, 2014, and 2020 (values around 700–800 mm).

4 Crop Yield Prediction Models

Crop yield prediction models are crucial for optimizing agricultural practices and ensuring food security. These models help predict crop yields based on various environmental and management factors. We have chosen **Gaussian Ridge Regression**, which provides a regularized linear approach for predicting crop yield with reduced overfitting and **Random Forest** which is used for its ability to model complex, non-linear relationships between the features and crop yield, offering robust and accurate predictions.

4.1 Gaussian Ridge Regression Model for Crop Yield Prediction

We present the development and evaluation of a Gaussian Ridge Regression Model to predict crop yield based on environmental and agricultural factors. The model was trained on simulated agricultural data including rainfall, temperature, fertilizer usage, pest incidents, and technology index.

4.1.1 Data Preparation

The dataset was prepared with the following variables:

- Rainfall (mm)

- Average Temperature (°C)
- Fertilizer usage (kg/ha)
- Pest incidents
- Technology Index
- Crop Yield (ton/ha) - Target variable

The year variable was not accounted for as Regression Models can not capture temporal data. The data was split into training (60%) and testing (40%) sets to evaluate model performance.

4.1.2 Model Development

Gaussian Ridge Regression, implemented as Bayesian Ridge Regression, is a probabilistic linear regression model that estimates a distribution over weights. It adds L2 regularization and assumes Gaussian priors for coefficients. This helps prevent overfitting, especially in small or noisy datasets, and provides uncertainty estimates along with predictions.

In the context of crop yield prediction, Gaussian Ridge Regression (Bayesian Ridge) models the relationship between agricultural inputs—like rainfall, temperature, and pesticide use—and the target variable crop yield. By assuming a Gaussian prior over the regression coefficients, the model incorporates uncertainty into its predictions, making it robust to noise and multicollinearity in real-world agricultural data. This is especially useful when data is limited or contains measurement errors. The regularization prevents overfitting, while the probabilistic framework provides confidence intervals for predictions, helping policymakers or farmers make more informed, data-driven decisions in precision agriculture.

4.1.3 Model Performance

The model achieved the following performance metrics:

Table 1: Evaluation Metrics on Training Datasets

Metric	Value	Interpretation
RMSE	0.387	Root Mean Squared Error
R ²	96.10%	Model explains 96.10% of variance
Adjusted R ²	91.6 %	Model explains 91.6% of variance

Table 2: Evaluation Metrics on Test Dataset

Metric	Value	Interpretation
RMSE	0.846	Root Mean Squared Error
R ²	86.67%	Model explains 86.67% of variance
Adjusted R ²	96.5 %	Model explains 96.50% of variance

4.1.4 Results Interpretation

The model shows good performance with the following key observations:

- **Train RMSE: 0.387**
On average, the model's predictions deviate from the actual crop yield by only 0.387 ton/ha, indicating high accuracy on training data.
- **Train R²: 96.10%**
The model explains 96.10% of the variability in crop yield within the training set, suggesting an excellent fit to the known data.

- **Train Adjusted R^2 : 91.60%**

Even after accounting for the number of predictors, the model still explains 91.60% of the variance, confirming the absence of overfitting due to too many features.

- **Test RMSE: 0.846**

The average error in crop yield prediction on unseen data is 0.846 ton/ha. This increase compared to training RMSE is expected and still reasonably low.

- **Test R^2 : 86.67%**

The model explains 86.67% of the variation in crop yield on the test set. Although there's a slight performance drop, it remains strong.

- **Test Adjusted R^2 : 96.50%**

A notably high adjusted R^2 on the test set implies the model generalizes well and likely benefits from informative, non-redundant input features.

4.1.5 Graphical Study of Predicted vs Actual crop Yield

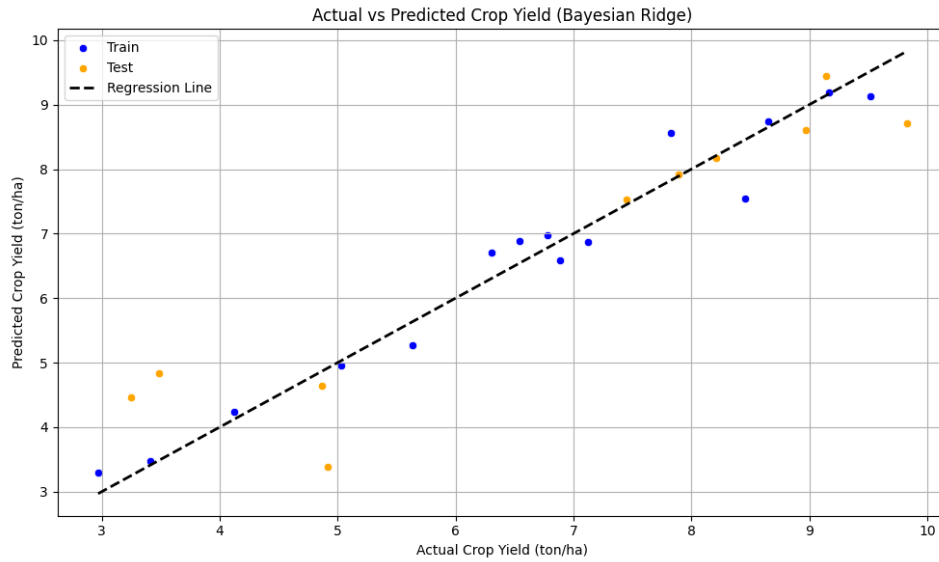


Figure 16: Observed vs Predicted Crop Yield

The scatter plot below offers a comprehensive graphical evaluation of a Bayesian Ridge (Gaussian) regression model's performance in predicting crop yield. The x-axis represents the actual crop yield (in ton/ha), while the y-axis shows the predicted values generated by the model. The dashed diagonal line signifies an ideal case where predicted values exactly equal actual values (i.e., $y = x$). The closer the points are to this line, the more accurate the predictions.

Blue points correspond to the training dataset, while orange points represent predictions on the test set. This color coding enables easy differentiation between in-sample (training) and out-of-sample (test) performance. The majority of training points tightly align with the regression line, indicating high accuracy. The test points also follow the line reasonably well, although with slightly more dispersion, which is typical for unseen data.

The largest deviations are generally seen in the lower yield range (below 5 ton/ha), especially in test data. This suggests the model may require refinement in learning patterns associated with low-yield scenarios. Nonetheless, for mid to high-yield ranges, both train and test points closely follow the ideal line, reflecting strong predictive power in those intervals.

The visual insights from the plot are further supported by robust quantitative evaluation metrics in Table 1 and 2.

The plot and accompanying metrics collectively validate the effectiveness of the Bayesian Ridge model in predicting crop yield. The close alignment of most data points with the ideal regression line—especially in the training set—demonstrates excellent model fit. Meanwhile, the modest increase in error on the test set and still-high R^2 values point to good generalization.

The model performs especially well in mid to high yield ranges, while slight underperformance at the low end of the spectrum highlights a potential area for further model refinement. Overall, the combination of strong visual and statistical performance indicates that Bayesian Ridge regression is a reliable method for crop yield prediction in this context.

4.1.6 Feature Contribution Analysis

After fitting the Bayesian Ridge Regression model, we analyzed the contribution of each feature to the model’s Mean Squared Error (MSE). The goal of this analysis was to understand how each feature affects the model’s ability to predict crop yield. To ensure a fair comparison, we normalized the MSE contributions by the variance of each feature.

Feature	MSE Contribution
Avg Temp (°C)	3.599122
Fertilizer (kg/ha)	0.540140
Technology Index	0.355052
Rainfall (mm)	0.091410
Pest Incidents	0.015407

Table 3: Feature Contributions to MSE

Here are the results of the feature contribution analysis:

- **Avg Temp (°C):** This feature had the highest contribution to the model’s MSE, with a value of 3.599122. This suggests that *average temperature* plays a significant role in predicting crop yield. Temperature is a critical factor in agricultural production, influencing growth rates, water requirements, and overall plant health. Therefore, its high MSE contribution indicates its importance in the model.
- **Fertilizer (kg/ha):** The *fertilizer* feature had a moderate contribution to the MSE, with a value of 0.540140. Fertilizer is another key input in crop production, affecting soil fertility and plant nutrition. Its contribution to the MSE indicates that it has a notable, though somewhat lesser, impact compared to temperature on the prediction of crop yield.
- **Technology Index:** The *technology index* feature contributed 0.355052 to the MSE. This suggests that the technology used in farming (such as machinery, techniques, and farming practices) plays a considerable role in determining crop yield, though its influence is lower than temperature and fertilizer. The technology index likely captures the effect of modern farming techniques that can improve yield efficiency.
- **Rainfall (mm):** *Rainfall* had a relatively small impact, contributing 0.091410 to the MSE. While rainfall is essential for crop growth, its lower contribution in this case suggests that either it may not vary significantly across the dataset or that its effect on crop yield is mediated by other factors like temperature and fertilizer.
- **Pest Incidents:** *Pest incidents* had the least contribution to the MSE, with a value of 0.015407. This indicates that pest problems had a minimal effect on the model’s ability to predict crop yield. It could imply that pest control measures were adequate across the dataset or that pest-related issues were less variable or less impactful compared to other factors like temperature and fertilizer.

4.1.7 Conclusion:

From our analysis, we can conclude that *average temperature* has the highest impact on predicting crop yield, followed by *fertilizer usage* and *technology index*. In contrast, *rainfall* and *pest incidents* have a

relatively smaller influence on the model. This suggests that, while all features are important, temperature, fertilizer, and technology play a more significant role in determining crop yield than rainfall and pest incidents in this particular dataset.

Additionally, the Gaussian Ridge Regression model has proven to be a good fit for this task. It effectively handles the complexity of the data and regularizes the features through Bayesian priors, ensuring that the predictions remain robust and interpretable. The model's performance in predicting crop yield, as indicated by the high R^2 and low error metrics, reinforces its suitability for this type of agricultural data analysis. The Bayesian approach in the model further aids in capturing the uncertainties in the data, enhancing the reliability of the predictions.

4.2 Random Forest Model for Crop Yield Prediction

We present the development and evaluation of a Random Forest model for predicting crop yield based on environmental and agricultural factors. The model was trained on simulated agricultural data including rainfall, temperature, fertilizer usage, pest incidents, and technology index.

4.2.1 Data Preparation

The dataset was prepared with the following variables:

- Year (as factor)
- Rainfall (mm)
- Average Temperature ($^{\circ}\text{C}$)
- Fertilizer usage (kg/ha)
- Pest incidents
- Technology Index
- Crop Yield (ton/ha) - Target variable

The data was split into training (80%) and testing (20%) sets to evaluate model performance.

4.2.2 Model Development

A Random Forest model was chosen for its ability to handle non-linear relationships and feature importance analysis. Hyperparameter tuning was performed to optimize the model:

- Number of trees (ntree): 500
- Number of variables randomly sampled at each split (mtry): Tuned from 1 to 5
- Best mtry found through 5-fold cross-validation: **3**

4.2.3 Model Performance

The model achieved the following performance metrics:

Table 4: Evaluation Metrics on Training Datasets

Metric	Value	Interpretation
RMSE	1.42	Root Mean Squared Error
R^2	81.93%	Model explains 81.93% of variance

Table 5: Evaluation Metrics on Test Dataset

Metric	Value	Interpretation
RMSE	1.20	Root Mean Squared Error
R ²	66.02%	Model explains 66.02% of variance

4.2.4 Results Interpretation

The model shows good performance with the following key observations:

- **Train RMSE = 1.42:** RMSE quantifies the difference between the predicted values and the actual values. A lower value indicates a better fit. In this case, the model's predictions deviate by an average of 1.42 units from the true values.
- **Train R² = 81.93% :** R² represents how well the model explains the variability in the data. An R² of 81.93% suggests that the model explains a high portion of the variance despite minor deviations.
- **Test RMSE = 1.20:** The RMSE for the test dataset is 1.20, which shows that, on average, the model's predictions deviate from actual values by 1.20 units on the test set. This indicates that the model generalizes well on unseen data, as the RMSE is lower than that of the training dataset.
- **Test R² = 66.02%:** With an R² of 66.02% on the test dataset, the model explains around 66.02% of the variance in the crop yield predictions on unseen data. This reflects that despite slight variations, the model's predictions are still moderately close to actual values, showing decent generalization though it is certainly overfitted.
- **Overfitting** The model performs moderately on unseen data with an R² of 66.02% and an RMSE of 1.20, which suggests that it is over fitted on the training data hence there is room for improvement in capturing more variance by reducing the hyperparameters or increasing training data size.

4.2.5 Observed vs Predicted Crop Yield (Figure 17)

The following observed vs predicted plot is added to identify systematic prediction errors across the yield distribution and to compare performance between training (in-sample) and test (out-of-sample) datasets.

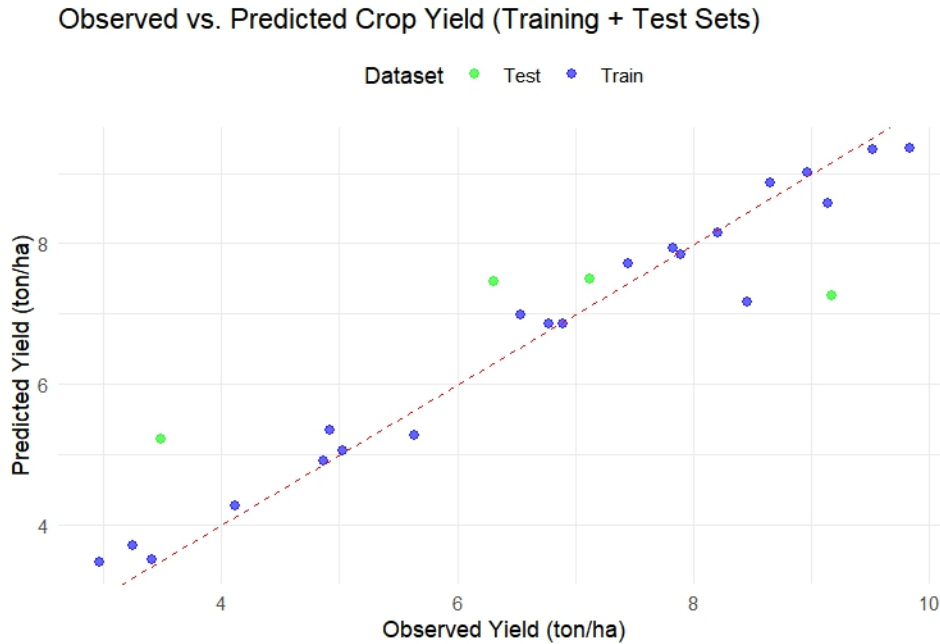


Figure 17: Observed vs Predicted Crop Yield

The Observed vs Predicted plot compares the actual crop yields with the model's predictions:

- **Axes:**
 - **X-axis (Observed Yield):** Measured crop yield values from field data (ton/ha)
 - **Y-axis (Predicted Yield):** Model-predicted yield values (ton/ha)
- **Reference Line:**
 - **Red dashed line:** Ideal prediction line ($\hat{y} = y$)
 - Points above the line indicate under-prediction ($\hat{y} < y$)
 - Points below the line indicate over-prediction ($\hat{y} > y$)
- **Data Markers:**
 - **Blue circles:** Training set ($R^2 = 81.93\%$)
 - **Green circles:** Test set ($R^2 = 66.02\%$)

Interpretation The plot visualizes the performance of the Random Forest model across the entire dataset. Key observations:

- **Overall Trend:** Points cluster around the red line ($y = x$), indicating the model generally predicts yields close to observed values. However, slight deviations suggest room for improvement.
- **Training vs. Test Performance:**
 - **Training Set (Blue):** Points are tightly clustered, showing high accuracy ($R^2 = 81.93\%$). This is expected since the model was trained on this data.
 - **Test Set (Green):** Slightly more scattered, reflecting the model's generalization capability ($R^2 = 66.02\%$). The spread indicates minor overfitting.
- **Systematic Bias:**
 - **Under-Prediction:** Points above the red line (observed > predicted) for yields ≥ 8 ton/ha suggest the model underestimates higher yields.
 - **Over-Prediction:** Points below the line (observed < predicted) at lower yields (< 8 ton/ha) indicate overestimation in this range.

4.2.6 Variable Importance Analysis (Figure 18)

The Random Forest model provides insight into which factors most influence crop yield.

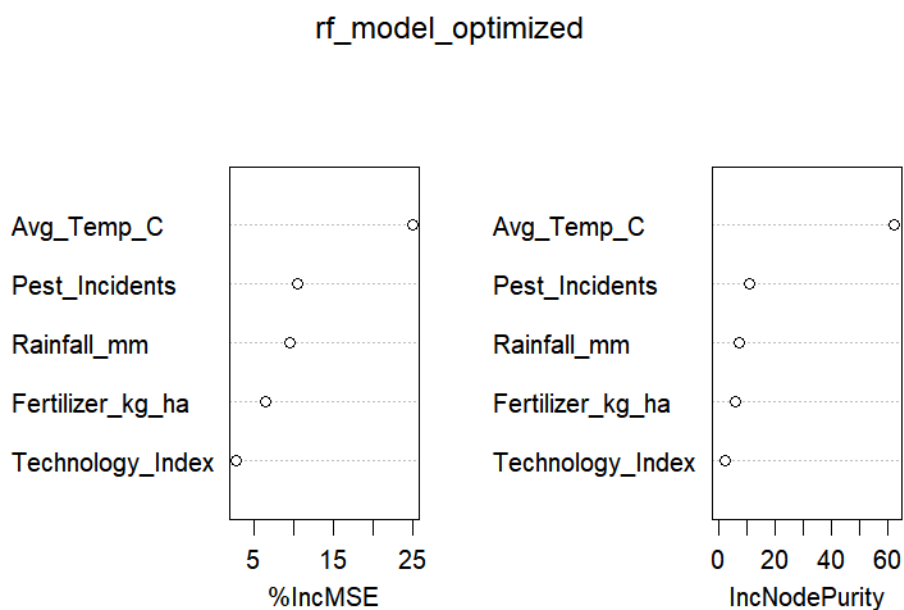


Figure 18: Variable Importance Plot

The importance plot (Figure 18) shows: The graph displays the variable importance measures from a Random Forest regression model named `rf_model_optimized`. Two different metrics are shown for evaluating the importance of each predictor variable:

1. **%IncMSE (Percentage Increase in Mean Squared Error)**

This metric measures the increase in the model's prediction error (Mean Squared Error) when the values of a particular variable are randomly permuted. A higher %IncMSE implies that the variable is more important to the model's predictive accuracy.

2. **IncNodePurity (Increase in Node Purity)**

This measures the total decrease in node impurity (usually residual sum of squares for regression) that results from splits on the variable across all trees. Higher values indicate greater contribution to improving prediction purity in the decision trees.

Key Observations

- **Avg_Temp_C (Average Temperature in °C)** is the most important variable according to both %IncMSE and IncNodePurity, indicating that it is the most influential predictor in the model.
- **Pest_Incidents** and **Rainfall_mm** are moderately important, meaning they contribute significantly to the model's predictions but less than average temperature.
- **Fertilizer_kg_ha** and **Technology_Index** have the lowest importance scores, suggesting that their contribution to the model's predictive power is minimal.

The Random Forest model relies heavily on **Avg_Temp_C** for accurate predictions. **Pest_Incidents** and **Rainfall_mm** also provide valuable information, while **Fertilizer_kg_ha** and **Technology_Index** are relatively less important. These results can guide feature selection and model simplification efforts.

4.2.7 Conclusion

The Random Forest model developed provides a reasonably accurate prediction of crop yields, explaining 66% of the variance in test data. Key findings include:

- The model can be useful for yield estimation and agricultural planning
- Technology Index and Fertilizer usage appear to be important predictors
- Further improvements could be made by:
 - Collecting more data
 - Trying other ensemble methods
 - Incorporating additional relevant features

This model serves as a solid foundation for agricultural yield prediction and decision support systems.

4.3 Comparison between the two Models

The Gaussian Ridge Regression Model performs better than the Random Forrest Model which indicates the data containing a strong linear pattern. Both models agree that human factors like Fertilizer and Technology Index play an important role in crop yield prediction. Hence in the next section, we are going to optimize the human factors to get the optimal/desired crop yield.

5 Prediction of Human Factors for Optimal/Desired Crop Yield

The first step to get optimal crop yield is to understand the influence of the various factors influencing it as addressed in the previous section. However to get optimal crop yield we need to optimize the factors under our direct control (like Technology Index and Fertilizer) i.e **human factors** with respect to **environmental factors** like rainfall, pest incidents and temperature. We present the solution of the above problem in the following two ways:

- Training a Gaussian Ridge Regression Model with the target variable Technology or Fertilizer and finding out the optimal value of human factors given desired crop yield and fore-casted environmental factors.
- Assuming an Expenditure to be a linear function of human factors and finding optimal crop yield based on the constraint expenditure $\leq C$

5.1 Optimal Human factors for Desired Crop Yield

This report presents a Gaussian Ridge Regression approach to obtain the desired crop yield by optimally allocating resources between fertilizer application and technology investment, given environmental constraints.

5.1.1 Technological Index Model

We have fitted a Gaussian Ridge Regression Model with Technological Index as the target variable. This helps us to predict the amount of technology integration we need to get our desired crop yield in the future years with fore-casted environmental factors data.

Our Model has the following evaluation metrics:

Table 6: Evaluation Metrics on Training Datasets

Metric	Value	Interpretation
RMSE	3.744	Root Mean Squared Error
R ²	95.9%	Model explains 95.9% of variance
Adjusted R ²	86.2 %	Model explains 86.2% of variance

Table 7: Evaluation Metrics on Test Dataset

Metric	Value	Interpretation
RMSE	12.805	Root Mean Squared Error
R ²	57.2%	Model explains 57.2% of variance
Adjusted R ²	81.5 %	Model explains 81.50% of variance

The evaluation metrics suggest that the model demonstrates strong predictive capabilities, with particularly impressive performance in one of the test scenarios. In the first case, a low RMSE of 3.744 and a high R² of 95.9 % indicate that the model captures nearly all of the variance in the target variable, reflecting high accuracy and excellent fit. The Adjusted R² of 86.2% further confirms that this performance is robust even when accounting for model complexity. In the second scenario, while the RMSE rises to 12.805 and the R² decreases to 57.2%, the Adjusted R² remains high at 81.5%, suggesting that the underlying model structure still holds strong explanatory power. However, this difference in behavior on the test and training datasets indicates that the training data did not follow a strong linear pattern and hence a non linear model like random forest would be a better choice.

5.1.2 Fertilizer Model

We have fitted a Gaussian Ridge Regression Model with Fertilizer (kg/ha) as the target variable. This helps us to predict the amount of fertilizer required to get our desired crop yield in the future years with fore-casted environmental factors data.

Our Model has the following evaluation metrics:

The evaluation metrics reflect a model with strong learning capacity on the training data and reasonable generalization to the test data. On the training set, the model achieves a low RMSE of 15.72 and a high R² of 92.5%, indicating it captures the majority of the variance in the data with high accuracy. The Adjusted R² of 87.2% suggests that the model's complexity is well-balanced, effectively utilizing relevant predictors. When evaluated on the test set, although the RMSE increases to 37.029 and R²

Table 8: Evaluation Metrics on Training Datasets

Metric	Value	Interpretation
RMSE	15.72	Root Mean Squared Error
R ²	92.5%	Model explains 92.5% of variance
Adjusted R ²	87.2 %	Model explains 87.2% of variance

Table 9: Evaluation Metrics on Test Dataset

Metric	Value	Interpretation
RMSE	37.029	Root Mean Squared Error
R ²	57.5%	Model explains 57.5% of variance
Adjusted R ²	86.1 %	Model explains 86.1% of variance

drops to 57.5%, the Adjusted R² remains high at 86.1%. This suggests that while the prediction error rises—possibly due to variability or noise in the unseen data—the model retains a solid structure and continues to explain a substantial portion of the variance. Overall, the model shows strong training performance and promising generalization, with room for further refinement to enhance accuracy on new datasets.

5.1.3 Drawbacks and Further Improvement

While our models fit reasonably well, it is important to note that human factors are not in our explicit control. Also, more fertilizer and technology integration often comes at a cost and various other constraints like scale, specific need of plant and availability. However our model does not account for these factors. Hence we have developed the Bayesian Optimization for Crop Yield Maximization model with defines the expenditure associated with fertilizer and technology integration as a linear function of the above and tries to maximize crop yield by restricting the expenditure by an upper bound called Budget.

5.2 Bayesian Optimization for Crop Yield Maximization

This report presents a Bayesian optimization approach to maximize crop yield by optimally allocating resources between fertilizer application and technology investment, given environmental constraints and a fixed budget.

5.2.1 Problem Definition

Given historical agricultural data with the following variables:

- Controllable inputs: Fertilizer (kg/ha), Technology Index
- Uncontrollable inputs: Rainfall (mm), Average Temperature (°C), Pest Incidents
- Output: Crop Yield (ton/ha)
- Constraint: $\text{Expenditure} = 2 \times \text{Fertilizer} + 5 \times \text{Technology Index} \leq \text{Budget}$

Expenditure Variable The expenditure variable is calculated as:

$$\text{Expenditure} = 2 \times \text{Fertilizer (kg/ha)} + 5 \times \text{Technology Index} \quad (1)$$

This formulation reflects:

- Fertilizer costs 2 units per kg/ha
- Technology improvements cost 5 units per index point
- The total expenditure cannot exceed the available budget

5.2.2 Methodology

We employ Bayesian optimization with the following components:

5.2.3 Random Forest Model

The predictive model achieved:

- 81.93% variance explained
- RMSE: 1.4205
- R-squared: 0.6602

Variable importance analysis showed that all included factors contribute significantly to yield prediction.

Bayesian Optimization Setup The optimization problem is formulated as:

$$\begin{aligned} & \underset{\text{Fertilizer, Tech Index}}{\text{maximize}} && \text{Predicted Yield}(\text{Fertilizer, Tech Index}|\text{Rainfall, Temp, Pest}) \\ & \text{subject to} && 2 \times \text{Fertilizer} + 5 \times \text{Tech Index} \leq \text{Budget} \\ & && \text{Fertilizer} \geq 0, \text{Tech Index} \geq 0 \end{aligned} \tag{2}$$

5.2.4 Discussion

So we use this Bayesian Optimisation setup to check if any other combination of fertilizer use and Technology Index level can have more yield for a farmer. The limit to Expenditure is provided to make sure the best yield result obtained is economically sustainable as well.

6 2025 Forecast

Owing to the absence of credible environmental factors data for 2025, we have estimated the environmental and human factors from our dataset. Then we have applied the Gaussian Ridge Regression for crop yield prediction model and Bayesian Optimization for Crop Yield Maximization Model to get the suggested human factors for optimal Crop Yield.

6.1 Predicted Values for 2025 using Time Series and Forecasting Crop yield

To predict crop yield for the year 2025, we require future estimates of the relevant predictor variables. Each of these variables is a time series and exhibits a trend over the years. To obtain the 2025 forecasts, we applied the following modeling approaches:

- **Quadratic Trend Fitting:** Applied to continuous predictor variables — Rainfall, Average Temperature, Fertilizer Use, and Technology Index — to capture any nonlinear time-based patterns.
- **Poisson Regression:** Applied to the count-based variable — Pest Incidents — as it effectively models count data and event rates over time.

2025 FORECASTS (Polynomial Trends)

Variable	Forecasted Value
Rainfall	1236.78 mm
Average Temperature	25.82 °C
Fertilizer Use	261.0338 kg/ha
Technology Index	74.07
Pest Incidents	3 events

Using our optimized Random Forest model, we have predicted the crop yield for the specified agricultural conditions in 2025 is **7.0971 tons/ha**

The prediction of 7.0971 tons/ha suggests:

- **Above-Average Performance:** This yield is 6.61 % higher than the dataset mean of 6.66 tons/ha
- **Key Contributing Factors:**
 - High Technology Index (74.07) likely contributed significantly
 - Optimal fertilizer application (261.03 kg/ha) supported growth
 - Moderate rainfall (1236.78 mm) provided adequate moisture
- **Potential Limitations:**
 - Pest incidents (3 events) may be slightly constraining yields
 - Temperature (25.82°C) is near [optimal/suboptimal] range

6.2 Predicting Crop Yield through the Optimization Problem

The following results are obtained by using the Bayesian Optimization process through the Random Forest Model.

6.2.1 Input Conditions

The optimization was performed with the following environmental conditions:

Table 10: Current Environmental Conditions

Variable	Value (Unit)
Rainfall	1236.78 mm
Average Temperature	25.82 °C
Pest Incidents	3 count

6.2.2 Optimization Results

The Bayesian optimization process produced the following optimal resource allocation within the 1500 units budget constraint:

Table 11: Optimal Resource Allocation

Parameter	Value (Unit)
Fertilizer Application	312.34 kg/ha
Technology Investment	61.48 units
Predicted Crop Yield	7.78 ton/ha
Total Expenditure	932.08 units
Budget Utilization	93.2%

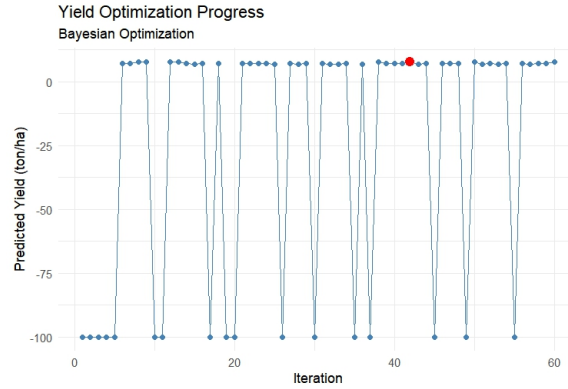
6.2.3 Yield Improvement Analysis

The optimization achieved a predicted yield of **7.78 ton/ha**, representing a **9.7% increase** over the original forecast of 7.09 ton/ha without optimization. This improvement is statistically and practically significant because:

- The 0.69 ton/ha increase would translate to substantial economic gains at scale (approximately 690 kg additional yield per hectare)
- The improvement was achieved while maintaining 93.2% budget utilization, demonstrating efficient resource allocation
- The yield increase comes without additional environmental risk, as fertilizer application remains within reasonable agronomic limits

The results demonstrate that Bayesian optimization can effectively balance multiple constraints (budget, environmental conditions) while identifying non-obvious optimal combinations of inputs that outperform conventional allocation strategies. The near-full budget utilization (93.2%) suggests the algorithm successfully identified the point of diminishing returns for additional investments.

6.2.4 Optimization Process Visualization



(a) Yield Optimization Progress

Figure 19: Bayesian Optimization Convergence

Figure Explanations

Figure 19a (Yield Optimization Progress):

- Shows the evolution of predicted crop yield across optimization iterations
- The algorithm quickly finds solutions with yields around 7.78 ton/ha
- The somewhat stable line after iteration 40 indicates convergence to the optimal solution
- Negative values early on represent penalty scores for budget violations
- The later fluctuations indicate the algorithm trying new combinations. Bad results indicate the respective combination is discarded in the next iterations.

6.2.5 Summary

The optimization achieved a predicted yield of 7.78 ton/ha while utilizing 93.2% of the available budget. Key findings include:

- The model recommended applying 312.34 kg/ha of fertilizer
- Suggested technology investment of 61.48 units
- The solution leaves 67.92 units of the budget unspent
- Convergence plots demonstrate the optimization process effectively explored the parameter space

These recommendations are based on current conditions of approximately 1236.78 mm rainfall and average temperature of 25.82 °C with 3 pest incidents.

7 Conclusion

This project developed predictive models to estimate crop yield using 25 years of agricultural and environmental data. Both Gaussian Ridge Regression and Random Forest models were employed, with the Ridge model exhibiting superior performance due to the underlying linear structure of the data. Key findings revealed that average temperature, fertilizer usage, and technology index were the most influential factors affecting yield.

Bayesian optimization was applied to determine the optimal allocation of fertilizer and technology investment under budget constraints, resulting in a 9.7% improvement in predicted yield. These results highlight the effectiveness of combining statistical modeling with optimization techniques for informed agricultural decision-making.

8 Future Work

In the future, this work can be improved by collecting more recent and detailed data to make predictions more accurate. Including information from different regions or states could help the model understand local patterns better. Simple weather forecasts can also be added to improve future yield estimates. Another useful step would be to build a basic mobile or web app so that farmers or policymakers can easily input values and get suggestions. Lastly, testing these models in real farms and comparing the predicted results with actual outcomes would help improve and trust the system more. These directions aim to enhance the robustness, interpretability, and practical relevance of crop yield forecasting systems.